PrivSearch: Expressive Privacy-Preserving Authenticated Search for Hybrid Blockchain Systems

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***Abstract*—Blockchain-assisted data management must reconcile privacy, verifiability, efficiency, and expressive search over outsourced data. This paper presents PrivSearch, a hybrid-blockchain framework supporting authenticated conjunctive, disjunctive, OR-of-AND, temporal-range, bounded-fuzzy, and semantic-v1 queries. PrivSearch decouples operational search from asynchronous blockchain anchoring and uses nonce-bound tokens to prevent replay. We prove correctness, completeness, replay resistance, latency separation, and simulation-based query privacy under explicit leakage. A composition theorem shows that one committed index state and a shared nonce registry support all six query families without family-specific cryptographic material. We also derive the exact soundness of the prototype's sampled accumulator verifier and identify its inadequate regime. A PureChain-integrated prototype using 2048-bit accumulator parameters and sample size $s = 5$ was evaluated on 834 workloads: 448 Enron-derived queries and 386 queries over 281,042 healthcare records. Mean authenticated retrieval latency was 5.25 ms for Boolean queries and 5.05 ms for fuzzy queries, with no verification failures under honest execution. All tested replay attempts were rejected, and 25 concurrent requests completed successfully. Relative to a same-stack SeaCQ baseline, verification work decreases by exactly $|R|/s$, with a corresponding single-omission detection probability of $s/|R|$.**

***Index Terms*—Authenticated search, cryptographic accumulators, hybrid blockchain, privacy-preserving search, replay protection, searchable encryption, verifiable query processing.**

I. Introduction

Healthcare, pharmaceutical supply chains, industrial Internet of Things (IIoT), finance, and digital governance increasingly depend on geographically distributed infrastructures that must simultaneously provide scalable storage, efficient retrieval, confidentiality, verifiable integrity, and regulatory compliance. Outsourcing sensitive data to semi-trusted providers introduces unauthorized disclosure, result manipulation, incomplete query responses, replay attacks, and dependence on untrusted operators, motivating blockchain-assisted data management, in which immutable ledgers establish decentralized trust while off-chain infrastructure provides scalable storage and computation [1]–[3].

Hybrid blockchain architectures have consequently become the predominant deployment model, maintaining bulk data off-chain while recording immutable metadata, integrity commitments, provenance, and access events on-chain [3]–[5]. This addresses scalability but does not guarantee trustworthy query processing: users must still rely on outsourced infrastructure to execute searches correctly, return complete results, and preserve confidentiality. The difficulty compounds because modern applications, clinical decision support, pharmaceutical traceability, digital forensics, regulatory auditing, extend well beyond exact-keyword search into conjunctive and disjunctive predicates, nested Boolean expressions, temporal filtering, approximate matching, and semantic retrieval, while requiring keyword privacy, correctness and completeness, response freshness, and defence against replay and query manipulation.

Progress has been substantial but fragmented. Authenticated query frameworks verify correctness and completeness of outsourced results [5]–[8] but support limited query semantics; searchable encryption enables confidential retrieval, from symmetric keyword search [9] to expressive Boolean queries [10] and blockchain-assisted healthcare applications [11]–[14], without efficient authenticated verification; hybrid blockchain databases prioritize storage scalability over expressive authenticated retrieval [3], [4]; and trusted execution approaches depend on specialized hardware [15]. Recent surveys accordingly identify authenticated retrieval, expressive query processing, privacy preservation, scalable verification, and practical deployment as persistent open challenges [1]. No existing framework provides all of expressive authenticated search, encrypted retrieval, replay protection, hybrid blockchain deployment, and formal security guarantees within one architecture.

We propose **PrivSearch**, a privacy-preserving authenticated search framework for hybrid blockchain systems. Unlike architectures that couple query execution tightly to blockchain operations, PrivSearch separates operational search from blockchain trust management: queries execute over authenticated encrypted indexes in the off-chain infrastructure, while blockchain commitments provide integrity anchors, provenance, and auditability. This separation preserves low operational latency without compromising security, and unifies authenticated conjunctive, disjunctive, OR-of-AND, temporal range, bounded fuzzy, and semantic retrieval within one framework alongside nonce-based replay protection.

The hybrid substrate on which PrivSearch is built, the PureChain anchoring plane, Merkle batch commitment, and the operational/anchoring latency-separation invariant, was introduced and evaluated for *plaintext* indexed logistics metadata in our earlier PureSearch framework [16], which does not address encrypted retrieval, query privacy, authenticated correctness or completeness, or replay resistance. That substrate

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is reused, not claimed; the contributions below concern the authenticated encrypted retrieval layer built above it.

The major contributions of this paper are summarized as follows.

- We design a unified authenticated search framework supporting six query families and nonce-bound replay protection while separating operational retrieval from asynchronous blockchain anchoring.
- We prove correctness, completeness, replay resistance, latency separation, and simulation-based query privacy under explicit leakage. A composition theorem establishes that one committed index state and a shared nonce registry support all query families under a common set of cryptographic assumptions.
- We derive the exact soundness of the prototype's sampled accumulator verifier and identify when sampling is inadequate. A controlled tamper-injection study evaluates this bound using 2048-bit accumulator parameters and the default sample size $s = 5$.
- We implement and evaluate PrivSearch on 834 workloads derived from Enron and a heterogeneous healthcare corpus containing 281,042 records. The evaluation covers latency, cryptographic overhead, scalability, replay rejection, tamper detection, concurrency, resource use, and distributed deployment.

Section II reviews related work and identifies the remaining gaps; Section III presents the PrivSearch framework, threat model, algorithms, and formal security analysis; Section IV describes the implementation, methodology, and evaluation; Section V concludes.

II. Related Work

Secure query processing over outsourced blockchain data spans four intersecting directions: efficient blockchain query processing, authenticated query services, searchable encryption, and hybrid or trusted-execution architectures [21]. Existing systems optimize only a subset of the requirements addressed by PrivSearch.

A. Query Processing, Authentication, and Encrypted Search

Blockchain storage is optimized for transaction validation rather than expressive retrieval, and several studies improve query efficiency through auxiliary indexes: index redesign for traceability queries [2], transaction-index mappings for manufacturing lifecycle retrieval [17], and hierarchical B-tree/LSM-tree combinations for equality and range queries [18]. VQL [19] separates query execution from consensus by querying a cloud database while anchoring integrity fingerprints on-chain, and query clustering [22] and common-table-expression optimization [23] address analytical workloads outside blockchain settings. These optimize retrieval under an honest-query-engine assumption, without encrypted search or client-side verification.

Authenticated data structures let clients verify correctness and completeness of outsourced results. EAQ [7] uses an MR Bloom-tree for authenticated multi-attribute range and object queries with proofs of nonexistence; the STK framework [6] extends authenticated querying to spatial–temporal–keyword and k-nearest-neighbor search; PAGB [8] combines RSA accumulators with completeness sets for graph queries. SeaCQ [5], the closest system to PrivSearch, supports encrypted authenticated conjunctive keyword search with two-stage verification and reduced on-chain overhead, but is limited to conjunctive predicates. More recent work reduces verification overhead or adds privacy-aware aggregation, MPV [24], EVQ [25], and VKFQ [26], without extending to disjunctive, temporal, fuzzy, or semantic retrieval. Authenticated query frameworks thus trade expressiveness for authentication efficiency.

Searchable encryption enables confidential retrieval over encrypted datasets, from symmetric keyword search [9] to expressive Boolean queries [10], forward-private verifiable conjunctive schemes [27], and misoperation-tolerant constructions [28]. Healthcare-specific extensions include blockchain-enabled key-aggregate searchable encryption [11], attribute-based searchable encryption with hidden policies [12], multi-source encrypted retrieval [13], verifiable conjunctive searchable encryption [14], and Healthchain [3]. These strengthen confidentiality and access control but emphasize keyword privacy over authenticated expressive query processing. A parallel line of work delimits what any leakage-bounded scheme can claim: access-pattern and query-recovery attacks [29], [30], file-injection attacks [31], inference attacks on property-preserving encryption [32], and volume-based reconstruction of range queries [33]. Section III-E states our position relative to these results explicitly.

B. Hybrid, Semantic, and Trusted Execution Approaches

Hybrid architectures store large datasets and indexes off-chain while preserving on-chain integrity commitments. MSTDB [4] employs a Merkle Semantic Trie for authenticated multi-keyword, range, Top-k, semantic, and cross-chain queries; Dutta et al. [20] combine metadata optimization with explainable AI for healthcare queries; TELEX [15] executes exact, Boolean, aggregate, and range queries inside secure enclaves, combining learned indexes with oblivious RAM to protect access patterns. These improve functionality and privacy but depend on trusted hardware or emphasize efficiency over encrypted authenticated retrieval. Asiamah et al. [1] identify authenticated retrieval, expressive query semantics, scalable verification, and privacy-preserving search as persistent open challenges.

C. Research Gap and Positioning of PrivSearch

Table I summarizes the capabilities of representative systems. Existing approaches exhibit four common limitations. First, indexing-oriented methods improve query efficiency but generally lack authenticated expressive retrieval [2], [17]–[19]. Second, authenticated query frameworks provide verifiable or privacy-aware results but remain specialized for particular workloads such as range, spatial–temporal, graph, or aggregate-frequency queries [6]–[8], [24]–[26]. Third, searchable encryption schemes, healthcare-specific and general-purpose alike, emphasize confidentiality, forward privacy, or robustness, but offer limited query expressiveness and little

TABLE I
COMPARISON OF PRIVSEARCH WITH REPRESENTATIVE SECURE AND BLOCKCHAIN QUERY FRAMEWORKS

Scheme	Block-chain	Hybrid Storage	Query Privacy	Result Verification	AND	OR	OR-of-AND	Range	Fuzzy	Semantic	Replay Protection	Formal Proofs	TEE Req.
Index-only [2], [17], [18]	✓	×	×	P	×	×	×	P	×	×	×	×	×
VQL [19]	✓	✓	×	✓	P	P	×	✓	×	×	×	P	×
BC-KASE [11]	✓	P	✓	P	P	×	×	×	×	×	✓	✓	×
ABSE-PH [12]	×	×	✓	×	×	×	×	×	×	×	×	✓	×
MOPSE [13]	×	×	✓	×	P	×	×	✓	×	×	×	✓	×
VC-KASE [14]	×	×	✓	✓	✓	×	×	×	×	×	×	✓	×
Healthchain [3]	✓	✓	✓	P	P	×	×	×	×	×	×	✓	×
EAQ [7]	✓	P	×	✓	P	×	×	✓	×	×	×	✓	×
STK Query [6]	✓	✓	×	✓	P	×	×	✓	×	×	×	✓	×
PAGB [8]	✓	✓	P	✓	×	×	×	×	×	×	✓	✓	×
SeaCQ [5]	✓	✓	✓	✓	✓	×	×	×	×	×	P	✓	×
MSTDB [4]	✓	✓	×	✓	P	×	×	✓	×	✓	×	P	×
XAI Metadata [20]	✓	P	P	×	×	×	×	×	×	P	×	×	×
TELEX [15]	✓	×	✓	✓	✓	✓	×	✓	×	×	×	P	✓
PrivSearch	✓	✓	✓	✓†	✓	✓	✓	✓	✓	✓	✓	✓	×

✓ = Supported; × = Not supported; P = Partially supported or not a primary design objective.
† Result verification is exhaustive under the complete verifier of Definition 1; the deployed prototype uses the sampled verifier, whose soundness is characterised in Theorem 3 and measured in Section IV-O.

support for blockchain-backed completeness verification [3], [11]–[14], [27], [28]. Finally, SeaCQ and TELEX improve either authenticated search or expressive querying but remain constrained by conjunctive query models or trusted hardware assumptions [5], [15].

PrivSearch addresses these limitations by means of a **security composition result** rather than by extending a single-family primitive. The OXT-style conjunctive primitive [9], [10] and SeaCQ's authenticated conjunctive path [5] each authenticate one query family against one committed index state. Section III-E establishes that a *single* shared committed state $\mathrm{Com}(I)$ simultaneously supports authenticated correctness and completeness across all six families, and that a *single* nonce registry enforces replay resistance across all request types, without separate commitment material or replay channels per family. This composition is not implied by any individual prior result. Its practical consequence is that PrivSearch deploys as a drop-in authenticated query layer over a hybrid blockchain stack without multiplying on-chain storage or cryptographic primitives by the number of supported query families.

III. SYSTEM METHODOLOGY

This section presents the PrivSearch architecture, adversary model, authenticated query framework, replay protection, formal security analysis, and analytical cost model.

A. Design Principles and System Architecture

PrivSearch preserves one architectural invariant: online query execution does not wait for blockchain anchoring. Let T_{search} and T_{anchor} denote search and asynchronous anchoring latency, respectively. Then

$$T_{search} \perp T_{anchor}, \tag{1}$$

where $\perp$ denotes separation of their critical paths rather than statistical independence. This invariant and the PureChain anchoring plane were established for plaintext indexed metadata in [16]; PrivSearch contributes the authenticated encrypted retrieval layer shown in Fig. 1.

Layer 1: Preparation and query API. During ingestion, the data owner cleans, normalizes, and de-identifies raw documents, computes their SHA-256 content hashes, encrypts payloads using AES-256-GCM, and constructs authenticated indexes and witnesses. During retrieval, the REST query API authenticates the data user, parses and routes the DNF query token, and applies a process-local nonce gate with validity window T (300 s in the prototype).

Layers 2–4: Index and encrypted storage. Layer 2 maintains two complementary indexes. The operational index stores non-sensitive metadata in a MongoDB B-tree with lookup complexity $O(\log n + |R|)$. The privacy index stores authenticated postings, HMAC-derived tokens, and the commitment $\mathrm{Com}(I)$; its OXT-style conjunctive search cost depends on the least-frequent posting list, $O(|R_{w_{\min}}|)$ [5]. Layer 3 stores AES-256-GCM-encrypted payloads in MinIO without persisting plaintext. Layer 4 optionally replicates encrypted objects through IPFS and records only their content identifiers in metadata.

Authenticated query and proof processing. The query processor routes conjunctive, disjunctive, OR-of-AND, temporal-range, bounded-fuzzy, and semantic-v1 queries to the execution engine. The engine retrieves authenticated postings, performs the required set operations or time-bucket lookups, and assembles the answer. The proof generator returns the answer A, proof P, commitment root, and index version, including the result digest, Merkle evidence, and sampled RSA-accumulator witnesses.

Off-chain commitment and audit manager. The manager constructs the versioned index commitment, computes Merkle roots over sealed audit batches, and places anchoring requests in an adaptive asynchronous queue. Its scheduler adjusts batch size according to λ/μ without placing anchoring on the read-query critical path.

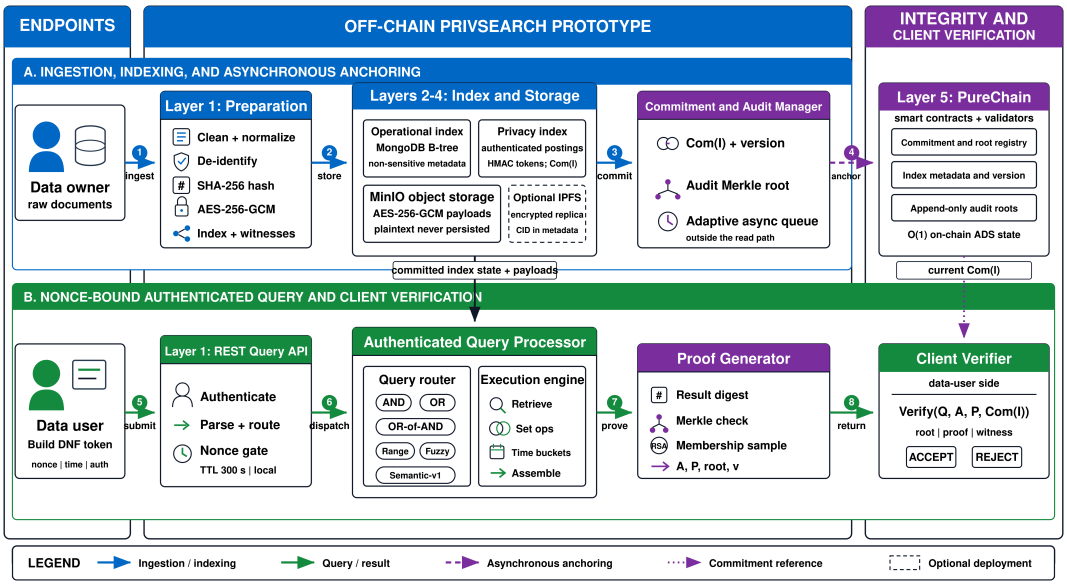


Fig. 1. PrivSearch architecture. Ingestion constructs authenticated encrypted indexes and asynchronously anchors their commitments on PureChain. Queries undergo nonce checking, authenticated search, proof generation, and client-side verification against the current commitment.

Layer 5: PureChain integrity. PureChain smart contracts and validators maintain the commitment-and-root registry, index metadata and version, append-only audit roots, and the constant-size on-chain ADS state. The current $\mathrm{Com}(I)$ is referenced by the data-user-side verifier, which evaluates

$$\mathrm{Verify}(Q, A, P, \mathrm{Com}(I))$$

and accepts or rejects the returned result.

B. System Model and Notation

Let $D = \{d_1, \ldots, d_n\}$ be the outsourced document set, W the keyword universe, $I(w)$ the authenticated posting list for $w \in W$, and $M(d)$ the authenticated metadata of document d. The current index commitment is denoted by $\mathrm{Com}(I)$. For query Q, answer A, and proof P, the client computes

$$\mathrm{Verify}(Q, A, P, \mathrm{Com}(I)) \in \{\mathrm{accept}, \mathrm{reject}\}. \tag{2}$$

After fuzzy or semantic expansion, every supported query is represented in disjunctive normal form:

$$Q = \bigvee_{i=1}^{d} C_i, \qquad C_i = \bigwedge_{j} (k_j = v_j) \wedge \bigwedge_{l} (r_l \in [a_l, b_l]), \tag{3}$$

and the final answer $A = \bigcup_i \sigma_{C_i}(D)$ is deduplicated by document identifier.

We assume honest data owners and users, an append-only blockchain, and a potentially malicious off-chain server that may fabricate, alter, omit, or replay information [5], [7], [8]. Byzantine validators and validator collusion are outside the retrieval-layer threat model because consensus security belongs to the reused PureChain substrate.

The analysis assumes DDH hardness for OXT cross-tags, pseudorandomness of HMAC-SHA256, authenticated-encryption security of AES-256-GCM under nonce uniqueness, and Strong RSA for the accumulator [34]. Experiments use a 2048-bit RSA modulus whose factorization was discarded after index preparation. One modular exponentiation costs 175.2 µs on Enron and 174.0 µs on the healthcare corpus. Production deployment requires either distributed modulus generation or a transparent accumulator, which remains outside the prototype scope. The prototype also integrates storage and query execution on the same stack; independent remote hosting remains future work.

TABLE II
THREAT MODEL AND MITIGATION

Adversary	Goal	Capability	Mitigation
Malicious server	Alter, omit, or fabricate results	Controls storage and query execution	Authenticated postings and committed index state; sampled soundness follows Theorem 3
Replay attacker	Repeat an accepted mutation	Replays valid request material	Nonce-bound freshness and exact-match registry
Stale-proof attacker	Return an outdated proof	Reuses a prior valid transcript	Equality check against current $\mathrm{Com}(I)$
Adaptive querier	Infer sensitive structure	Selects queries from prior outputs	Encrypted postings and leakage-bounded trapdoors

C. Authenticated Query Framework

PrivSearch supports six query families through authenticated posting-list intersection and union.

1) Conjunctive Query: For $Q_{and} = w_1 \wedge \cdots \wedge w_k$,

$$A_{and} = \bigcap_{j=1}^{k} I(w_j). \tag{4}$$

This OXT-style path is also used for the same-stack SeaCQ comparison in Section IV-G.

2) Disjunctive Query: Each OR clause is evaluated and verified independently:

$$Q_{or} = \bigvee_{i=1}^{m} c_i, \qquad A_{or} = \bigcup_{i=1}^{m} A(c_i), \quad A(c_i) = \bigcap_{w \in c_i} I(w). \tag{5}$$

The client rejects if any clause fails verification and otherwise returns the deduplicated union. Thus, OR support requires no additional trusted component.

3) OR-of-AND Query: General DNF queries are evaluated as

$$Q_{oa} = \bigvee_{i=1}^{p} \left(\bigwedge_{j=1}^{q_i} w_{ij} \right), \qquad A_{oa} = \bigcup_{i=1}^{p} \bigcap_{j=1}^{q_i} I(w_{ij}). \tag{6}$$

Each conjunctive clause uses the AND engine, followed by verified deduplication.

4) Temporal-Range Query: Let $\mathcal{B}(a, b)$ be the authenticated metadata buckets covered by $[a, b]$. Then

$$A_{range} = \left(\bigcup_{t \in \mathcal{B}(a,b)} M^{-1}(t) \right) \cap \bigcap_{j} I(k_j). \tag{7}$$

Each bucket is converted to a keyed token in the same namespace as keyword terms; literal bucket values are neither stored nor transmitted. A range query is therefore a union of $B(a, b) = |\mathcal{B}(a, b)|$ authenticated lookups and requires no order-preserving encryption or its associated order leakage [32].

5) Bounded Fuzzy Query: For edit distance $\mathrm{ed}(\cdot, \cdot)$, PrivSearch supports threshold-one matching:

$$Q_{\text{fuzzy}}(w) = \{w' \in W : \mathrm{ed}(w, w') \le 1\}, \qquad A_{\text{fuzzy}} = \bigcup_{w' \in Q_{\text{fuzzy}}(w)} I(w'). \tag{8}$$

This covers one insertion, deletion, or substitution, not arbitrary-distance similarity search.

6) Semantic Query (v1): A concept c is mapped to a curated expansion set $E(c)$ with client-known weights ω_t:

$$A_{sem} = \bigcup_{t \in E(c)} I(t), \qquad \mathrm{score}(d) = \sum_{\substack{t \in E(c) \\ d \in I(t)}} \omega_t. \tag{9}$$

Results are ranked by $\mathrm{score}(d)$ and verified against their authenticated postings. This executable semantic-v1 path performs concept expansion, not embedding-based retrieval.

D. Replay-Protection Mechanism

Each state-mutating request carries $(nonce, ts, auth)$. The bounded, TTL-evicting registry $\mathcal{N}$ uses the key

$$\kappa = (principal, method, path, nonce), \tag{10}$$

which binds freshness to a principal and endpoint. Admission is

$$\mathrm{Admit}(\kappa) = \begin{cases} \text{reject}, & \kappa \in \mathcal{N}, \\ \text{insert } \kappa;\ \text{process}, & \text{otherwise}. \end{cases} \tag{11}$$

The exact-match check precedes mutation and therefore rejects a repeated request while its key remains resident. This improves on the post-hoc replay detection of the reused substrate [16]. The guarantee requires sufficient capacity C for all live keys within validity window T. Moreover, the prototype registry is process-local and neither survives restart nor coordinates API replicas; horizontally scaled deployment requires a shared freshness store.

E. Formal Security Analysis

Definition 1 (Complete and sampled verifiers)**.** *The complete verifier validates all authenticated membership and coverage evidence needed to reconstruct the committed answer. The sampled verifier checks membership witnesses for a uniformly selected subset of* $\min(s, |A|)$ *returned documents. They coincide when* $|A| \le s$*. The prototype uses* $s = 5$*.*

Theorem 1 (Conjunctive correctness)**.** *Under the complete verifier, acceptance implies that every returned document satisfies all conjunctive terms in the committed index state.*

Proof. Let $d \in A_{and}$. By Eq. (4), d is returned only if it appears in every posting list $I(w_j)$. The complete verifier authenticates each contributing posting against $\mathrm{Com}(I)$. If $d \notin I(w_j)$ for any j, the corresponding membership evidence is invalid and verification rejects. Therefore, acceptance implies $d \in \bigcap_{j=1}^{k} I(w_j)$, so every returned document satisfies all query terms. □

Theorem 2 (Conjunctive completeness)**.** *Under the complete verifier, acceptance implies that no valid committed conjunctive result is omitted.*

Proof. Let

$$R_{and} = \bigcap_{j=1}^{k} I(w_j)$$

be the result induced by the committed postings. The complete verifier validates the authenticated coverage evidence for every contributing posting and reconstructs their deterministic intersection. If some $d^* \in R_{and}$ were omitted from A_{and}, then $A_{and} \neq R_{and}$, causing the reconstructed result or its committed coverage digest to disagree with the submitted proof. The verifier would reject, contradicting acceptance. Hence $A_{and} = R_{and}$. □

The preceding theorems apply to the complete verifier. They do not imply equivalent completeness for the prototype's sampled verifier.

Theorem 3 (Sampled-verifier soundness)**.** *Let a returned set contain* n *documents, including* $t \ge 1$ *invalid documents. If* $s \le n$ *documents are sampled uniformly without replacement, then*

$$p_{\det}(n, t, s) = 1 - \frac{\binom{n-t}{s}}{\binom{n}{s}} = 1 - \prod_{i=0}^{t-1} \frac{n - s - i}{n - i}, \tag{12}$$

with $p_{\det} = 1$ *when* $s > n - t$*. For* $t = 1$*,* $p_{\det} = s/n$*.*

Proof. Detection fails exactly when all s checked documents are selected from the $n - t$ valid documents. There are $\binom{n-t}{s}$ such samples among $\binom{n}{s}$ equiprobable samples. Therefore, the failure probability is their ratio and the detection probability is its complement. Expanding the ratio yields the product form. For $t = 1$, the failure probability reduces to $(n - s)/n$, giving $p_{\det} = s/n$. □

Corollary 1 (Required sampling rate)**.** *Detecting one invalid document with probability at least $1-\delta$ requires $s \geq (1-\delta)n$. Hence, fixed-size sampling provides no result-size-independent detection floor.*

Proof. For $t = 1$, Theorem 3 gives $p_{\text{det}} = s/n$. Requiring $s/n \geq 1 - \delta$ yields $s \geq (1 - \delta)n$. If s is fixed while n increases, then $s/n \to 0$. □

Corollary 2 (Repeated interaction)**.** *For q independent tampered responses with per-query detection probability p, the probability of at least one detection is $1 - (1 - p)^q$.*

Proof. The probability that all q tampered responses avoid detection is $(1 - p)^q$. Taking the complement gives the stated result. □

Remark 1 (No omission completeness under sampling). Because sampling selects only returned documents, an omitted document cannot be sampled. Therefore, the sampled verifier detects invalid inclusions probabilistically but does not establish completeness.

Accordingly, zero verification failures under an honest server demonstrate functional correctness, not adversarial soundness. Tamper detection is evaluated separately in Section IV-O.

Theorem 4 (OR-query correctness and completeness)**.** *Let $Q_{or} = c_1 \vee \cdots \vee c_m$. If every clause answer is accepted under the complete verifier and the final answer is their deduplicated union, then the returned answer equals the committed OR-query result.*

Proof. Let $R(c_i)$ denote the committed result of clause c_i. By Theorems 1 and 2, acceptance of each clause gives $A(c_i) = R(c_i)$. Therefore,

$$A_{or} = \bigcup_{i=1}^{m} A(c_i) = \bigcup_{i=1}^{m} R(c_i).$$

Deduplication removes only repeated identifiers and neither introduces an invalid document nor removes a distinct valid document. Hence A_{or} is exactly the committed OR-query result. □

Corollary 3 (OR verification cost)**.** *OR verification cost is additive across independently authenticated clauses.*

Proof. Each clause is evaluated and verified independently before union. The total verification work is therefore the sum of the clause-verification costs plus deduplication of their returned identifiers. □

Theorem 5 (Range-query correctness and completeness)**.** *Under the complete verifier, an accepted range answer equals the union of the committed buckets covered by $[a, b]$, intersected with any accompanying equality predicates.*

Proof. Let

$$R_{range} = \left(\bigcup_{t \in \mathcal{B}(a,b)} M^{-1}(t) \right) \cap \bigcap_{j} I(k_j).$$

Every returned document must possess authenticated metadata placing it in a covered bucket and must satisfy each accompanying equality predicate. Thus, an out-of-range or predicate-violating document causes membership verification to fail. Conversely, the complete verifier authenticates the coverage of every bucket in $\mathcal{B}(a, b)$ and the participating equality postings. Omitting a document from their deterministic union and intersection makes the submitted answer inconsistent with the authenticated coverage evidence. Therefore, acceptance implies $A_{range} = R_{range}$. □

Corollary 4 (Range-query cost)**.** *Range-query cost increases with bucket count $B(a, b)$ and candidate-set size $|R_{range}|$.*

Proof. The engine performs one authenticated lookup for each bucket in $\mathcal{B}(a, b)$, followed by union, optional intersection, and verification over the resulting candidates. Hence work increases with both $B(a, b)$ and $|R_{range}|$. □

Theorem 6 (Replay resistance under residency)**.** *Assume that no live key is evicted from $\mathcal{N}$ during freshness window T. A request inducing key κ cannot cause two accepted state mutations within T.*

Proof. The first accepted request inserts κ before executing its mutation. Every subsequent request inducing the same key is checked against $\mathcal{N}$ before mutation. Under the residency assumption, κ remains present and the exact-match lookup rejects the request. Therefore, the same freshness tuple cannot produce a second accepted mutation within T. □

Corollary 5 (Capacity condition)**.** *For peak accepted-request rate λ, residency is guaranteed when $C \geq \lambda T$. If $C < \lambda T$, admissible requests may evict a live key and invalidate the guarantee.*

Proof. At most λT distinct live keys arrive during a window of length T. Capacity $C \geq \lambda T$ therefore retains all live keys. If $C < \lambda T$, an adversary that generates sufficiently many admissible keys can force eviction before expiry, after which the evicted request may be replayed. □

Replay checking has expected $O(1)$ hash-table cost.

Theorem 7 (Latency separation)**.** *Because the authenticated read path does not synchronously invoke anchoring, its protocol latency is independent of anchoring backlog, except for incidental contention over shared resources.*

Proof. The query path comprises normalization, trapdoor generation, off-chain retrieval, proof generation, and client verification. Anchoring executes through a separate asynchronous queue and contributes no waiting term to this critical path. Increasing the anchoring backlog therefore does not change protocol-level read latency. Shared CPU, memory, disk, or network contention may affect observed latency, but such interference is an implementation effect rather than a protocol dependency. □

This restates Eq. (1) for encrypted retrieval; the underlying anchoring separation was established in [16].

Definition 2 (PrivSearch leakage)**.** *PrivSearch retains the OXT trapdoor, TSet, and cross-tag construction [10]. Its setup leakage is*

$$\mathcal{L}_{setup}(DB) = (|DB|, N), \qquad N = \sum_{w \in W} |I(w)|,$$

and its query leakage is

$$\mathcal{L}_{qry}(Q_t \mid Q_{<t}) = \big(\mathrm{sp}(Q_t), \mathrm{ap}(Q_t), |R_{Q_t}|, \mathrm{XP}(Q_t), \mathrm{IP}(Q_t)\big),$$

where sp, ap, XP, *and* IP *denote the search, access, x-term equality, and conditional intersection patterns, respectively. Union-based queries additionally reveal their clause count.*

Theorem 8 (Simulation-based query privacy)**.** *Under DDH, secure pseudorandomness of F, and secure authenticated encryption, PrivSearch is $\mathcal{L}$-secure in the random-oracle model:*

$$|\Pr[\mathrm{Real}_{\Pi,\mathcal{A}}(\lambda) = 1] - \Pr[\mathrm{Ideal}_{\Pi,\mathcal{A},\mathcal{S}}(\lambda) = 1]| \le \mathrm{negl}(\lambda).$$

Proof sketch. The conjunctive core retains the OXT trapdoor, TSet, and cross-tag constructions without modification. Under DDH and pseudorandomness of F, the OXT simulator constructs an indistinguishable encrypted-index and query transcript using only $\mathcal{L}_{setup}$ and $\mathcal{L}_{qry}$. Secure authenticated encryption hides payload contents, while the accumulator can be simulated from the identifier set already exposed through the permitted access pattern. The union extension follows from Theorem 9. The complete real/ideal experiments and hybrid sequence are given in the supplementary material. □

Theorem 9 (Union composition)**.** *If Π is $\mathcal{L}$-secure, independently evaluating m clauses and deduplicating their answers client-side is secure under*

$$\mathcal{L}'(\mathbf{q}) = \big(m, \mathcal{L}(c_1), \ldots, \mathcal{L}(c_m)\big).$$

Proof. Consider hybrids $H_0, \ldots, H_m$, where H_j replaces the first j real clause transcripts with simulated transcripts. Adjacent hybrids differ in one invocation of Π, so their distinguishing advantage is negligible by the $\mathcal{L}$-security of Π. Summing over m hybrids gives at most $m \cdot \mathrm{negl}(\lambda)$, which remains negligible for polynomial m. Deduplication occurs client-side after decryption and therefore reveals no additional server-side overlap information. Hence the composed scheme is $\mathcal{L}'$-secure. □

Remark 2 (Temporal-range coverage). A range query comprises $B(a, b)$ independently trapdoored bucket lookups and is therefore covered by Theorem 9. Token shape does not distinguish bucket from equality lookups.

The privacy result bounds leakage but does not eliminate access-pattern attacks, including query recovery [29], [30], file injection [31], or volume-based range reconstruction [33]. The static prototype also provides neither forward nor backward privacy; dynamic schemes such as [27] remain complementary. The accumulator adds no information beyond the committed identifier set because its public witnesses can be simulated from that set.

Theorem 10 (Cross-family security composition)**.** *Let*

$$\mathcal{Q} = \{Q_{\mathrm{and}}, Q_{\mathrm{or}}, Q_{\mathrm{oa}}, Q_{\mathrm{range}}, Q_{\mathrm{fuzzy}}, Q_{\mathrm{sem}}\}$$

share commitment $\mathrm{Com}(I)$ *and nonce registry $\mathcal{N}$. Under Strong RSA and the assumptions of Theorems 6 and 8, the following properties hold without family-specific commitment material:*

(i) *accepted answers satisfy their authenticated query predicates; forging membership for an uncommitted identifier violates Strong RSA;*
(ii) *replay resistance applies uniformly to protected mutations, subject to nonce residency and the endpoint-bound key of Eq. (10);*
(iii) *the complete verifier preserves correctness and completeness across families, whereas the sampled verifier provides only the guarantee of Theorem 3; and*
(iv) *adaptive multi-family transcripts remain simulatable under the leakage of Definition 2, extended by clause counts according to Theorem 9.*

Proof sketch. All query families reduce to authenticated intersections or unions of postings committed by the same $\mathrm{Com}(I)$. Conjunctive correctness and completeness follow from Theorems 1 and 2; OR, OR-of-AND, fuzzy, semantic-v1, and range queries preserve these properties through authenticated union by Theorems 4, 5, and 9. A forged membership witness for an uncommitted identifier contradicts Strong RSA. Replay resistance follows from the common pre-mutation registry check of Theorem 6. Finally, Theorem 8 simulates the shared conjunctive core, while Theorem 9 extends the simulation to union-based and range families. The sampled implementation inherits only Theorem 3 and not complete omission detection. The full hybrid composition is provided in the supplementary material. □

Thus, one authenticated index commitment and one replay-protection channel serve all six query families.

F. Analytical Cost Model

End-to-end query latency is decomposed as

$$\begin{aligned} T_{query}(Q) = {} & T_{norm}(Q) + T_{trapdoor}(Q) + T_{retrieve}(Q) \\ & + T_{proof}(Q) + T_{verify}(Q) + T_{anchor_ref}(Q), \end{aligned} \tag{13}$$

where T_{anchor_ref} measures commitment-reference checking, not asynchronous anchoring.

For a k-term conjunctive query verifying sample R_s,

$$T_{verify}(Q_{and}) = O(k + |R_s| t_{exp}). \tag{14}$$

For an m-clause OR query with deduplicated union U,

$$T_{query}(Q_{or}) = \sum_{i=1}^{m} T_{query}(c_i) + O\left(\sum_{i=1}^{m} |A(c_i)|\right), \tag{15}$$

and

$$T_{verify}(Q_{or}) = \sum_{i=1}^{m} T_{verify}(c_i) + O(|U|). \tag{16}$$

For a temporal range,

$$T_{query}(Q_{range}) = O(B(a, b) + |R_{range}| + |R_s| t_{exp}). \tag{17}$$

Let $S_{index} = S_{enc} + S_{meta}$. Storage overhead is

$$\rho_{storage} = \frac{S_{index}}{S_{raw}}, \tag{18}$$

and proof size is

$$S_{proof} = O(k b_{root} + |R_s| b_{wit}). \tag{19}$$

TABLE III
PRIVSEARCH COMPLEXITY

Operation	Complexity	Driver
Metadata search	$O(\log n + \lvert R\rvert)$	Result size
Conjunctive search	$O(\lvert R_{w_{\min}}\rvert)$	Smallest posting
Disjunctive search	$O(\sum_i \lvert R^{(i)}_{w_{\min}}\rvert)$	Clause postings
Range search	$O(B + \lvert R_{range}\rvert)$	Buckets and results
Complete verification	$O(\lvert R\rvert k t_{exp})$	All witnesses
Sampled verification	$O(\min(s, \lvert R\rvert) k t_{exp})$	Sample size
Returned-set digest	$O(\lvert R\rvert t_h)$	All returned records
Witness precomputation	$O(\sum_w \lvert I(w)\rvert^2 t_{exp})$	Posting-list sizes
Merkle batching	$O(b)$	Batch size
Blockchain anchor	$O(1)$	Fixed root
Replay lookup	Expected $O(1)$	Exact-match hash
On-chain ADS	$O(1)$	Accumulator value

At fixed s, sampled witness checking is constant in $|R|$, but returned-set hashing remains linear. Sampling therefore reduces accumulator work without establishing omission completeness or eliminating result-size-dependent verification cost.

IV. IMPLEMENTATION, EXPERIMENTAL METHODOLOGY, AND PERFORMANCE EVALUATION

This section describes the prototype, experimental protocol, performance results, security evaluation, and deployment limitations.

A. Implementation and Reproducibility

PrivSearch is implemented as a runnable prototype (`privsearch-prototype`) integrated with the PureChain-backed stack of [16]. It implements HMAC-SHA256 trapdoors, AES-256-GCM encrypted postings, the six query families of Section III-C, Merkle-style integrity checks, and sampled RSA-accumulator verification. MongoDB provides the operational B-tree index, MinIO stores encrypted payloads, PureChain receives asynchronous commitments, and the nonce registry operates in-process at the REST API.

Table IV summarises all hardware, software, and statistical configuration needed to reproduce the reported measurements. MongoDB, MinIO, PureChain, and the PrivSearch REST server all execute on the same host; an independently hosted query server remains future work.

Dataset versions and retrieval dates. The **Enron Email Corpus** [35] (517,401 emails, 150 mailboxes) was obtained from the CMU Machine Learning Repository on 2026-06-15. Body text was extracted, lowercased, whitespace-tokenized, and stripped of a 11-token noise set (e.g., *ect*, *hou*, *corp*); date fields were parsed as ISO-8601 timestamps.

The **eICU Collaborative Research Database demo v2.0.1** [36] (MIT Laboratory for Computational Physiology, retrieved 2026-06-15) contributed patient, diagnosis, and medication records from 200 hospitals; 100,582 records were retained after schema normalization. The **MedCD Clinical Document Dataset** [37] (TigerResearch, 2025; retrieved 2026-06-15) is a Chinese-language structured corpus of admission, discharge, lab-test, and imaging records; 1,999 records were ingested from the `medcd-annotations` split (`medcd_search_notes.csv` and `medcd_search_queries.csv`). Because MedCD text is in Chinese, keyword extraction used a mixed-language tokenizer: English tokens were whitespace-tokenized and stop-word filtered, while Chinese tokens were extracted as overlapping 2–3-character Han n-grams via Unicode `Script=Han` segmentation. The **openFDA drug enforcement, NDC directory, and recall records** [38] were retrieved via the openFDA REST API on 2026-06-15; 178,461 documents were retained across all three endpoints. All corpora were lowercased, whitespace-tokenized, and deduplicated within each source before keyword extraction; no cross-corpus deduplication was performed. The merged healthcare corpus contains 281,042 records and 221,760 unique keywords.

TABLE IV
REPRODUCIBILITY CONFIGURATION

Parameter	Value
Hardware	
CPU	AMD Ryzen 7 7700 (8 cores / 16 threads, 3.8 GHz base)
RAM	32 GB DDR5
Storage	NVMe SSD (Crucial CT1000T500SSD8, 1 TB PCIe)
OS	Windows 11 Pro 64-bit, build 26200
Software	
Node.js	v24.15.0 (built with OpenSSL 3.5.0)
Crypto library	Node.js `crypto`: HMAC-SHA256, AES-256-GCM; custom JS RSA mod-exp kernel
MongoDB	v7.0.31 (Docker image `mongo:7`)
MinIO	RELEASE.2025-09-07T16-13-09Z (Docker)
Docker	v29.4.0 / Docker Compose v5.1.1
PureChain	External EVM (chain ID 900520900520); Solidity 0.8.24, Hardhat 2.22.19
TypeScript / Python	5.4.3 / 3.11.9
ethers.js	v6.13.5
Measurement conditions	
CPU freq. scaling	Not controlled (Windows Balanced power plan)
Measurement threads	Sequential; one HTTP request per trial; no background concurrent load
Statistical parameters	
Bootstrap CIs	3,000 resamples; seeds 314,159 (flat) and 271,828 (hierarchical)
Permutation tests	10,000 resamples, two-sided; seed 42
Tamper trials	100 per (t, s, bucket) cell; OS-entropy via `crypto.randomBytes()`

Workload generation. Workload-generation methodology is described in Section IV-B; complete manifests are in the project repository.

Benchmark commands. Primary latency benchmarks ran as:

```
node dist/benchmark/run.js --dataset enron \
  --engine privsearch-prototype --trials 100
node dist/benchmark/run.js --dataset healthcare \
  --engine privsearch-prototype --trials 100
```

The SeaCQ comparison, tamper injection, concurrency, and resource-overhead benchmarks are invoked by the corresponding scripts in `dist/benchmark/`. Figures are generated by

`scripts/generate-privsearch-ieee-pub-figures.py`.

Repository and artifacts. Benchmark scripts, the SeaCQ baseline, generated result assets (`Results/paper_assets/`), and raw benchmark logs are retained in the project repository. A persistent URL and DOI will be provided upon acceptance.

Supplementary proofs. The full hybrid-argument proof for Theorem 8 (simulation-based query privacy) and the cross-family composition proof for Theorem 10 are provided in the supplementary material submitted with this revision.

Cryptographic parameters and preparation cost. The accumulator uses a 2048-bit RSA modulus. One modular exponentiation costs 175.2 µs on Enron and 174.0 µs on the healthcare corpus. Membership witnesses are generated once during index preparation and cached for both PrivSearch and the SeaCQ baseline. Per-witness generation averaged 836.0 ms for Enron posting lists containing 3,113 entries and 16.80 s for healthcare lists containing 65,864 entries, reaching 168 s for the largest lists. Because the current procedure scales quadratically with posting-list length, witness generation is the primary deployment constraint; batched or incremental updates remain future work.

The prototype recomputes clauses from decrypted postings but samples accumulator witnesses. It therefore realizes the probabilistic verifier of Theorem 3, not the complete verifier of Theorems 1–5. It also lacks on-chain query-specific digests and a generalized arbitrary-range engine.

B. Datasets and Workloads

The **Enron-derived track** [35] contains 448 workloads: 436 single-keyword, Boolean, temporal-range, and bounded-fuzzy workloads, plus 12 semantic-v1 workloads. Each workload was executed for 100 measured trials.

The **healthcare track** combines eICU demo [36] diagnosis and medication records, MedCD notes [37], and openFDA [38] enforcement, NDC, and recall-product records. After normalization, it contains 281,042 records: 100,582 from eICU, 1,999 from MedCD, and 178,461 from openFDA. Its 221,760 unique keywords support 386 workloads across the six query families, also evaluated over 100 trials. Non-empty conjunctions for $k \in \{6, 8, 10\}$ were obtained by exhaustive intersection over the 120 most frequent keywords. Together, the tracks comprise 834 workloads.

Workload-generation methodology. Keywords were stratified into four document-frequency buckets: rare (5–20 docs), low (21–100), medium (101–1,000), and high (1,001–15,000). Noise-like tokens (e.g., *net*, *www*, *corp*) and strings outside the character-length range 4–24 or containing non-alphabetic characters were excluded; 30–50 candidates were sampled evenly from each bucket. *Conjunctive workloads* drew keywords from the medium and high buckets only. At k=2, all pairwise intersections with $|R| \geq 10$ were enumerated and a representative subset selected by even spacing. At $k \in \{4, 5, 6, 8, 10\}$, a greedy overlap-targeting algorithm seeded each query with one medium/high keyword, then added the candidate whose intersection best approximated a target overlap ratio of 35% (k=2) to 70% (k=10), subject to $|R| \geq 10$ at every extension step. *Disjunctive workloads*: OR-2 enumerated all pairwise unions across all four buckets; OR-3 enumerated all triples from an even per-bucket subsample; OR-of-AND workloads paired disjoint k=2 conjunctions whose keyword sets were non-overlapping. *Range workloads* enumerated every contiguous calendar window of width $\{1, 3, 6\}$ months and $\{2, 3, 5\}$ years lying within the Enron 1999–2002 date span, requiring at least 20 emails per month window; representative subsets were selected by even spacing. *Fuzzy workloads* used 6 seeds from each of the four frequency buckets (24 seeds total). Each fuzzy query term was formed by deleting the penultimate character of the seed keyword, yielding a Levenshtein-1 variant; the query resolves at runtime to the original seed's authenticated posting list. *Semantic workloads* matched 12 predefined domain concepts (energy contracts, power generation, trading risk, finance, legal compliance, travel logistics, etc.) against the indexed vocabulary. Each workload returns the union of all matched keyword postings, capped at K=10 results. Healthcare workloads used the same procedure; non-empty conjunctions at $k \in \{6, 8, 10\}$ required exhaustive intersection over the 120 most frequent healthcare keywords. Complete workload manifests (`enron_query_workloads/` and `healthcare_query_workloads/`) are included in the project repository.

C. Experimental and Statistical Protocol

We report mean, median, SD, bootstrap 95% CI (3,000 resamples), P95, and P99 latency. Between-family comparisons use two-sided permutation tests with 10,000 resamples and Cliff's-δ effect sizes over workload-level means (see Table IV for seeds). Direct experimental comparison is limited to SeaCQ, reimplemented on the same stack in Section IV-G. TELEX [15], EAQ [7], VQL [19], and MSTDB [4] are compared only at the feature level. Embedding-based retrieval, independent remote query hosting, and geographically distributed deployment remain outside the experimental scope.

D. Enron-Derived Query Results

Table V summarizes the 436 non-semantic workloads. All honest-server trials were accepted without verification failures.

Across the 244 Boolean and fuzzy workloads, mean latency was 5.2503 ms, with verification contributing 97.69%. Conjunctive latency increased from 2.59 ms at $k = 2$ to 12.90 ms at $k = 10$, consistent with the $O(\min(s, |R|)kt_{exp})$ model. Under the tested clause structures, OR and OR-of-AND latency remained comparable to conjunctive execution.

Broad ranges and high-cardinality keyword queries were substantially slower. The two-year range averaged 1,329.78 ms, including 1,093.80 ms of verification, because its mean result set contained 166,209 records. Sampling bounds accumulator checks but does not remove result-set assembly and hashing costs that remain linear in $|R|$. Accordingly, the low-latency claim applies to Boolean, fuzzy, semantic-v1, and bounded-range workloads, not broad temporal scans. The non-monotonic range results reflect result cardinality rather than interval width, as predicted by Eq. (17).

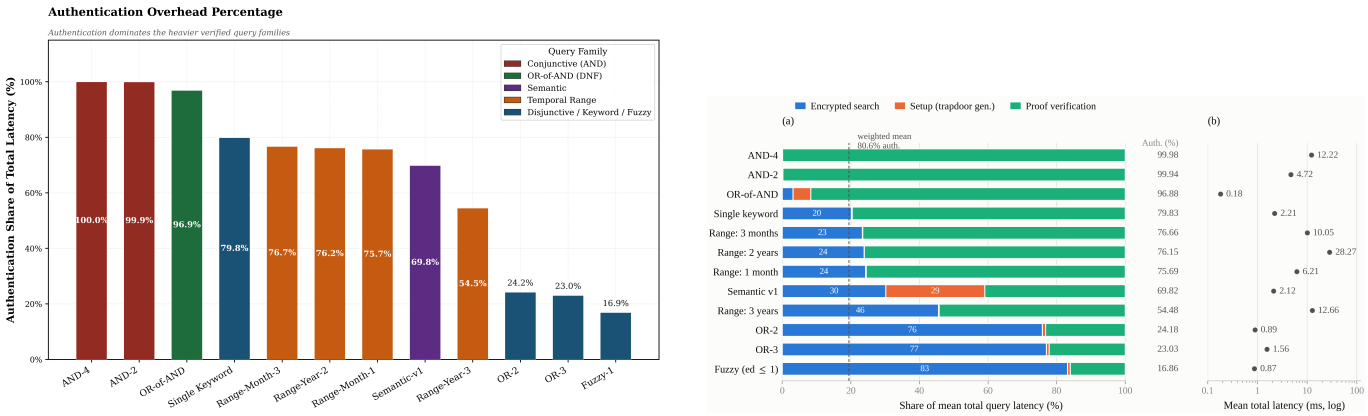


Fig. 2. Authentication share of total latency on the Enron-derived (left) and healthcare (right) tracks.

TABLE V
ENRON-DERIVED QUERY-FAMILY RESULTS

Query Type	Count	Total (ms)	Search (ms)	Verify (ms)	Fail.
single_keyword	150	17.9511	0.1609	17.7845	0
and_2	40	2.5898	0.0453	2.5374	0
and_4	20	5.1399	0.0690	5.0582	0
and_5	20	6.4817	0.0793	6.3835	0
and_6	20	7.7866	0.0968	7.6699	0
and_8	20	10.3369	0.1237	10.1886	0
and_10	20	12.8967	0.1428	12.7234	0
or_2	40	2.6389	0.1398	2.4913	0
or_3	20	4.1290	0.2368	3.8772	0
or_of_and	20	5.2231	0.1230	5.0860	0
fuzzy_1	24	1.3356	0.0542	1.2770	0
range_month_1	12	179.6280	30.5292	149.0893	0
range_month_3	10	375.7152	64.3724	311.3173	0
range_month_6	8	262.1934	47.5155	214.6375	0
range_year_2	6	1329.7822	235.9553	1093.7975	0
range_year_3	4	35.4835	29.4649	5.9758	0
range_year_5	2	62.0726	52.2419	9.7756	0

TABLE VI
SELECTED HEALTHCARE QUERY-FAMILY MEANS

Query Type	Count	Total (ms)	Search (ms)	Verify (ms)
single_keyword	240	40.5753	2.5299	38.0396
and_2	24	3.2350	0.4235	2.8017
and_4	12	5.3278	0.1541	5.1575
and_6	6	7.8838	0.1821	7.6803
and_8	6	10.3833	0.2641	10.0932
and_10	6	13.0665	0.3526	12.6799
or_2	24	3.3355	0.7852	2.5404
or_3	12	5.5967	1.6782	3.9013
or_of_and	12	6.7653	1.0907	5.6546
fuzzy_1	12	1.4450	0.1719	1.2672
semantic_v1	12	12.6985	0.6275	4.3540
range_month_1	8	81.1254	13.6651	67.4517
range_month_3	6	144.6576	24.5747	120.0662
range_year_2	4	410.9589	70.5322	340.4091
range_year_3	2	135.9195	25.5169	110.3744

E. Semantic-v1 Results

Semantic-v1 was evaluated on 12 concept-expanded workloads per corpus. Mean Enron latency was 755.8416 ms: 315.8987 ms for expansion setup, 71.2563 ms for search, and 368.6867 ms for verification. Healthcare latency was 12.6985 ms: 7.7169 ms for setup, 0.6275 ms for search, and 4.3540 ms for verification. The difference is driven by expansion breadth and posting-list length. These results characterize authenticated concept expansion over keyword postings, not embedding-based semantic retrieval.

F. Healthcare-Corpus Validation

All 386 healthcare workloads completed without honest-server verification failures. The workload-weighted mean was 36.0062 ms, comprising 3.3046 ms of search and 32.4530 ms of verification. This aggregate is influenced by 240 single-keyword workloads; the 114 Boolean and fuzzy workloads averaged 5.0466 ms, close to the Enron result of 5.2503 ms.

The similar Boolean costs across corpora indicate that k and s, rather than total corpus size, dominate selective queries. In contrast, healthcare single-keyword and range queries are heavier because their posting lists are longer. Verification contributes 90.13% of healthcare latency, consistent with Fig. 2.

G. Same-Stack SeaCQ Comparison

SeaCQ's conjunctive path was reimplemented using the same host, Node.js runtime, inverted indexes, warm-up protocol, and cached witnesses. The implementations share the OXT-style search path but differ in verification: SeaCQ checks every returned document in every contributing list, whereas PrivSearch checks at most $s = 5$. Their accumulator work is therefore

$$\frac{\text{SeaCQ operations}}{\text{PrivSearch operations}} = \frac{|R|k}{\min(s, |R|)k} = \frac{|R|}{s}, \qquad |R| > s. \tag{20}$$

The operation-count ratio is independent of k and hardware. Enron SeaCQ latency was measured directly; healthcare SeaCQ latency was derived from exact operation counts and the measured $t_{exp} = 174.0$ µs.

Enron produced a stable 2.0–2.1× reduction because $|\bar{R}| \approx 10$. Healthcare ratios ranged from 12.1× to 184.5× because its result sets were larger. PrivSearch verification remained approximately linear in k but bounded in $|R|$; when $|R| \leq s$, both approaches perform the same accumulator work.

This efficiency has a corresponding soundness cost. For the healthcare $k = 10$ row, checking 5 of 918.7 results reduces accumulator work by 183.7× but detects a single invalid result

TABLE VII
SAME-STACK CONJUNCTIVE COMPARISON

Corpus	k	$\lvert\bar{R}\rvert$	PS ops	SQ ops	Ops ratio	PS Ver. (ms)	SQ Ver. (ms)
Enron	2	10.3	10	20.7	2.1×	2.446	5.215
Enron	4	10.3	20	41.3	2.1×	5.110	10.374
Enron	6	10.0	30	60.0	2.0×	7.717	15.261
Enron	8	10.0	40	80.0	2.0×	10.043	19.812
Enron	10	10.0	50	100.0	2.0×	12.487	24.805
Health	2	102.0	10	204.0	20.4×	3.106	35.504‡
Health	4	60.3	20	241.3	12.1×	5.497	42.002‡
Health	6	379.7	30	2278.0	75.9×	8.544	396.463‡
Health	8	922.7	40	7381.3	184.5×	11.292	1284.647‡
Health	10	918.7	50	9186.7	183.7×	13.680	1598.847‡

PS: PrivSearch with $s = 5$; SQ: exhaustive SeaCQ-style verification. Operations are 2048-bit modular exponentiations. Enron SQ latency was measured; ‡healthcare SQ latency was derived as $\lvert\bar{R}\rvert k t_{exp}$. Witnesses were precomputed for both systems.

TABLE VIII
SELECTED STATISTICAL COMPARISONS

Comparison	p-value	Cliff's δ	Sig.
Enron `single_keyword` vs. `and_2`	0.0029	+0.223	yes
Enron `and_2` vs. `and_10`	< 0.0002	−1.000	yes
Enron `and_2` vs. `or_2`	0.3105	−0.045	no
Healthcare `and_2` vs. `or_2`	0.9453	+0.313	no
Healthcare `semantic_v1` vs. `fuzzy_1`	< 0.0002	+0.986	yes
Enron `and_2` vs. Healthcare `and_2`	< 0.0002	−0.606	yes

with probability only $5/918.7 \approx 0.54\%$. Thus, Eq. (20) and Theorem 3 describe opposite sides of the same trade-off.

H. Literature-Based Comparison

Table I provides the broader comparison. TELEX supports rich private queries but assumes trusted hardware; EAQ provides authenticated multiattribute retrieval without query-content privacy; VQL verifies off-chain queries without privacy-preserving search; and MSTDB supports hybrid semantic queries without authenticated private retrieval. Because these systems were not reproduced on the same stack, no numerical performance superiority is claimed.

I. Statistical Significance

Latency distributions for high-cardinality workloads were strongly right-skewed. For example, Enron `single_keyword` had a mean of 17.9511 ms and median of 3.6704 ms, while `range_year_2` had a mean of 1,329.7822 ms and median of 383.6428 ms. We therefore use bootstrap rather than normal-approximation intervals. Boolean workloads were substantially less skewed because their sampled-verification cost is primarily governed by k and s.

The $k = 2$ versus $k = 10$ comparison showed complete separation ($\delta = -1.000$), supporting linear growth with clause count. By contrast, AND and OR did not differ significantly on either corpus, consistent with their similar accumulator-operation counts. Healthcare $k = 2$ conjunctions were modestly but significantly slower than their Enron counterparts. Statistical power is determined by the number of independent workloads, not the 100 repeated trials; small family-level effects should therefore be interpreted cautiously.

TABLE IX
GLOBAL DEPLOYMENT-COST SUMMARY

Metric	Value
Prepared posting sets	102
Indexed document references	1,279,790
Raw postings bytes	37,164,996
Encrypted payload bytes	74,339,172
Encrypted index size bytes	74,434,355
Blockchain commitment bytes	74,101
Storage overhead $\rho_{storage}$	2.0028×

J. Ablation Study

The ablation progressed from plaintext postings (A0) through encrypted retrieval (A1), authenticated conjunction (A2), OR/OR-of-AND (A3), range processing (A4), replay protection (A5), and the integrated system (A6/A6b). Encryption increased latency from 0.0232 to 1.0731 ms, whereas authentication increased it to 100.5184 ms, including 99.4453 ms of verification. OR support added 3.7714 ms. The range stage reached 3,893.8788 ms because of its large result set, while replay checking added only 0.0205 ms for acceptance and 0.0036 ms for rejection. The ablation used 10 trials per query in one session and supports relative component attribution; Tables V and VI remain authoritative for absolute performance.

K. Resource and Storage Overhead

Resource measurements cover 53 Enron-side workloads with three trials each.

The encrypted index approximately doubles raw posting storage, while blockchain commitments occupy only 9.95×10^{-4} of the encrypted-index size. Mean proof size ranges from 966 bytes for single-keyword and range queries to 7,216 bytes at $k = 10$, growing with clause count but remaining bounded by s. CPU and memory values are process-level deltas; proof, commitment-reference, and network sizes are serialized-payload estimates rather than packet-capture measurements.

L. Cryptographic Overhead

Across all 448 Enron workloads, cryptographic operations contributed 95.68% of mean latency. Proof verification alone accounted for 48.1446 ms, or 73.62% of the 65.3992 ms workload-weighted total. On healthcare, the cryptographic and proof-verification shares were 99.30% and 90.13%, respectively. Trapdoor generation and proof construction remained below 0.06 ms per family; encrypted-posting retrieval became substantial only for range and semantic-v1 queries. Inline commitment-reference checking contributed no measurable query-path latency, consistent with Theorem 7.

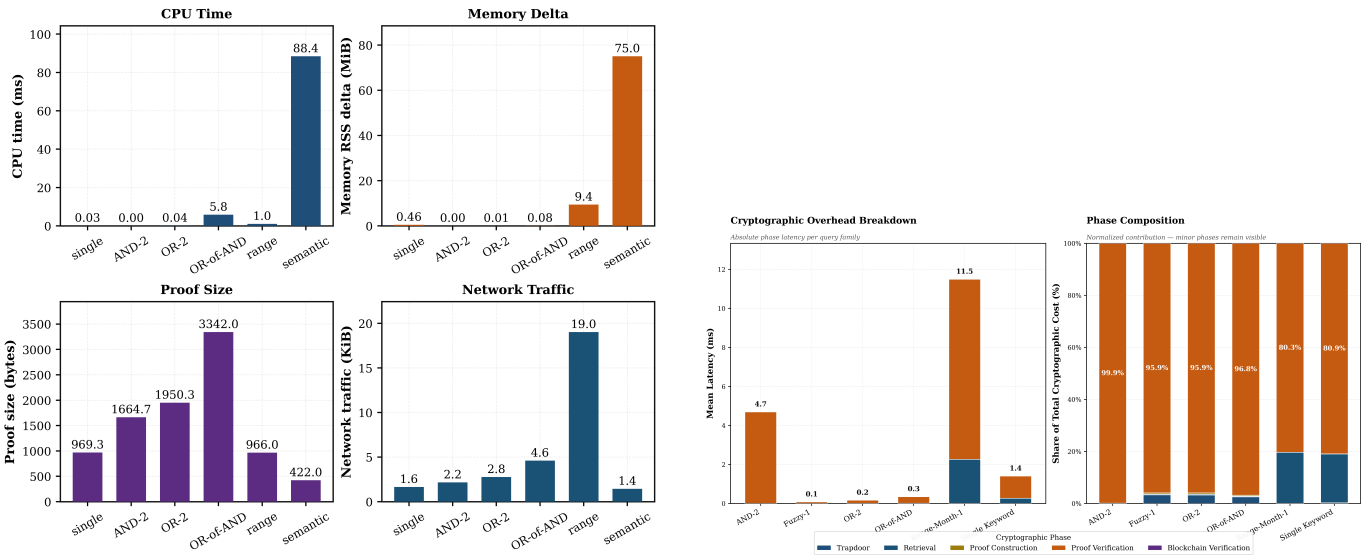


Fig. 3. Resource overhead by query family (left) and latency decomposition by pipeline stage (right).

M. Scalability

Posting-list size. Mean latency increased from 2.6901 ms for lists of 1–10 documents to 4,829.4158 ms above 10,001 documents. Verification contributed 4,003.51 ms in the largest bucket, confirming that result-set processing remains linear in $|R|$. Proof size remained nearly constant at 970–982 bytes because only s witnesses were transmitted.

Clause count. Conjunctive latency increased from 2.5961 ms at two clauses to 13.6511 ms at ten clauses, approximately 1.38 ms per additional clause. Proof size increased from 1,670 to 7,212 bytes. Disjunctive and DNF workloads followed the same trend.

Concurrency. Concurrency levels of 1, 5, 10, and 25 achieved 100% success without measured queueing or latency inflation. For the representative authenticated conjunctive family (AND-2), throughput remained approximately 379–400 req/s, indicating stable but primarily serialized verification rather than multicore scaling. Simpler query families (e.g., range queries) achieve higher per-server throughput; AND-2 represents the typical authenticated-search workload. Horizontal scaling requires multi-process workers and a shared nonce registry.

N. Replay and Stale-Proof Protection

Across 200 live-API tests, PrivSearch accepted all first requests and rejected all 200 replays. Mean replay-rejection latency was 0.7897 ms (P95: 1.1734 ms), compared with 581.4113 ms for an accepted authenticated write. The in-process registry check accounted for 0.0036 ms; the remaining rejection time arose from HTTP handling. The experiment used required mode, a 300 s TTL and clock-skew allowance, and 200 resident keys. Capacity-pressure and restart behavior were not evaluated; the guarantee remains subject to Corollary 5.

Each proof also contains its index-state root. Before witness verification, the client compares this root with the current $\mathrm{Com}(I)$; a proof from a superseded state therefore fails immediately. This stale-proof mechanism is implemented but was not evaluated through a dedicated injection experiment.

TABLE X

EMPIRICAL TAMPER-DETECTION RATES AND CORRECTED THEORETICAL PROBABILITY $\bar{p}_{\mathrm{det}}$: 11 REPRESENTATIVE CELLS FROM THE 48-CELL (t, s) × BUCKET MATRIX (N=100 TRIALS EACH). $\bar{p}_{\mathrm{det}} = \frac{1}{N}\sum_i p_{\mathrm{det}}(n_i, t, s)$ AVERAGES OVER THE N ACTUAL WORKLOAD RESULT SIZES n_i IN EACH BUCKET, REPLACING THE PRIOR PRACTICE OF EVALUATING ONLY AT THE BUCKET MEAN $\bar{R}$. EMPIRICAL 95 % CIS USE THE WILSON INTERVAL.

t	s	Bucket	$\bar{R}$	Empirical [95 % CI]	$\bar{p}_{\mathrm{det}}$
1	5	small	20	40% [31, 50]	35.1%
1	10	small	20	68% [58, 76]	62.7%
1	25	small	20	95% [89, 98]	92.3%
5	5	small	21	76% [67, 83]	80.9%
10	5	small	24	95% [89, 98]	92.8%
1	5	medium	167	4% [2, 10]	4.3%
5	25	medium	167	62% [52, 71]	64.0%
10	50	medium	167	95% [89, 98]	94.6%
1	5	large	3603	0% [0, 4]	0.3%
1	50	large	3603	2% [1, 7]	3.0%
10	50	large	3603	22% [15, 31]	24.2%

O. Tamper-Detection Efficacy

We substituted $t \in \{1, 2, 5, 10\}$ fabricated identifiers into legitimate result sets and sampled $s \in \{5, 10, 25, 50\}$ returned documents. Each (t, s) configuration was evaluated over 100 trials in small, medium, and large result-set strata. Detection occurs when at least one fabricated identifier is sampled.

Across the complete 48-cell matrix, the mean absolute difference between empirical and $\bar{p}_{\mathrm{det}}$ is 2.2 percentage points, down from 4.0 pp when theory was evaluated only at $\bar{R}$. The correction is most visible in the small-bucket row where s=25, $\bar{R}$=20: evaluating at the mean gave $p_{\mathrm{det}}(20, 1, 25)$=100 % (because $s > \bar{R}$), yet the empirical rate was 95 % and $\bar{p}_{\mathrm{det}}$=92.3 %. Of the 140 small-bucket workloads, 39 (28 %) have n_i>25, so sampling of size s=25 no longer exhausts the answer and detection falls below

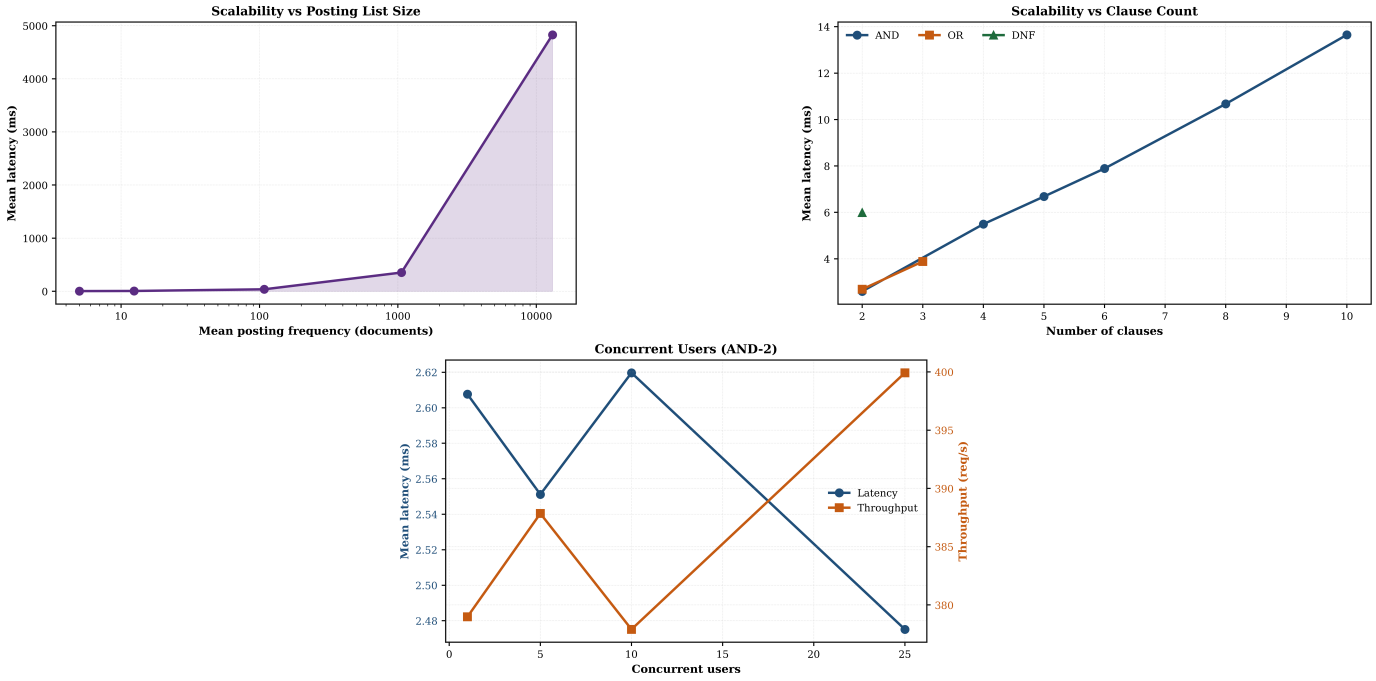


Fig. 4. Scalability with posting-list size (left), clause count (centre), and concurrency (right, AND-2 representative).

certainty. For workloads where $|R| \leq s$, sampling is exhaustive and detection is guaranteed.

The critical limitation appears on large result sets. At $\bar{R} \approx 3{,}603$, the default $s = 5$ detected a single fabricated result in none of 100 trials, consistent with the theoretical probability of approximately 0.1%. Even $s = 50$ detected only 2%. Hence, fixed-size sampling provides no meaningful single-query protection against isolated fabrication in large answers. Achieving detection floor α requires $s \approx \alpha|R|$, surrendering the sublinear work reduction.

Repeated interaction provides cumulative auditing rather than per-query soundness: detection probability after q independent tampered responses is $1 - (1 - p)^q$. Exhaustive verification remains the $s = |R|$ endpoint, and sampling does not detect omissions as stated in Remark 1.

P. WAN-Style Distributed Validation

Local HTTP validator peers were evaluated under injected LAN, regional-WAN, and cross-site-WAN delay and jitter for 2, 4, 7, and 10 nodes. All trials, each using a 50-transaction candidate block and 51% threshold, succeeded. At 10 cross-site nodes, threshold collection averaged 94.5550 ms versus 125.1294 ms for all-node collection. Signature verification remained below 2.5 ms, making validation network-bound. Because the experiment used local machines with injected delay rather than geographically separated hosts, it provides WAN-style latency evidence but not multi-site deployment evidence or Byzantine-validator validation.

Q. Discussion

The evaluation establishes that all six query families and real-time replay rejection are executable across 834 workloads. Selective Boolean and fuzzy queries averaged 5.25 ms on Enron and 5.05 ms on healthcare, while broad range and high-cardinality searches remained result-size bound. Authentication dominated latency, whereas asynchronous anchoring contributed no measurable query-path term.

The SeaCQ comparison exposes the principal trade-off: accumulator work decreases by $|R|/s$, while single-fabrication detection decreases to $s/|R|$. The default $s = 5$ is therefore suitable only for low-cost auditing or small result sets, not strong per-query integrity for large answers. Deployments requiring deterministic correctness and completeness must use exhaustive verification or stronger authenticated coverage mechanisms. Other limitations include quadratic witness preparation, a process-local nonce registry, static-index leakage, serialized verification, and the absence of geographically distributed query-server evaluation.

V. Conclusion

This paper presented PrivSearch, a privacy-preserving authenticated search framework that separates off-chain query execution from asynchronous blockchain anchoring. The framework supports conjunctive, disjunctive, OR-of-AND, temporal-range, bounded-fuzzy, and semantic-v1 queries with nonce-based replay protection. Formal analysis establishes correctness and completeness for the complete verifier, probabilistic soundness for the sampled verifier, replay resistance under nonce residency, latency separation, and simulation-based privacy under explicit leakage.

The prototype was evaluated at 2048-bit accumulator parameters on 834 workloads, including 386 workloads over 281,042 healthcare records. Boolean and fuzzy queries averaged 5.25 ms on Enron and 5.05 ms on healthcare, with no verification failures under honest execution. All 200 tested replays were rejected, and all requests succeeded at concurrency levels up to 25. Relative to a same-stack SeaCQ baseline, sampled

verification reduces accumulator work by exactly $|R|/s$, but detects a single invalid result with probability only $s/|R|$.

This trade-off defines the principal limitation: fixed-size sampling provides weak per-query protection for large results and no omission completeness, while result processing remains linear in $|R|$. Other limitations include quadratic witness preparation, a process-local nonce registry, and concept-expansion rather than embedding-based semantic retrieval. Future work will investigate dynamic accumulators, adaptive or exhaustive verification, authenticated coverage proofs, shared freshness storage, embedding-based retrieval, and geographically distributed deployment.

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